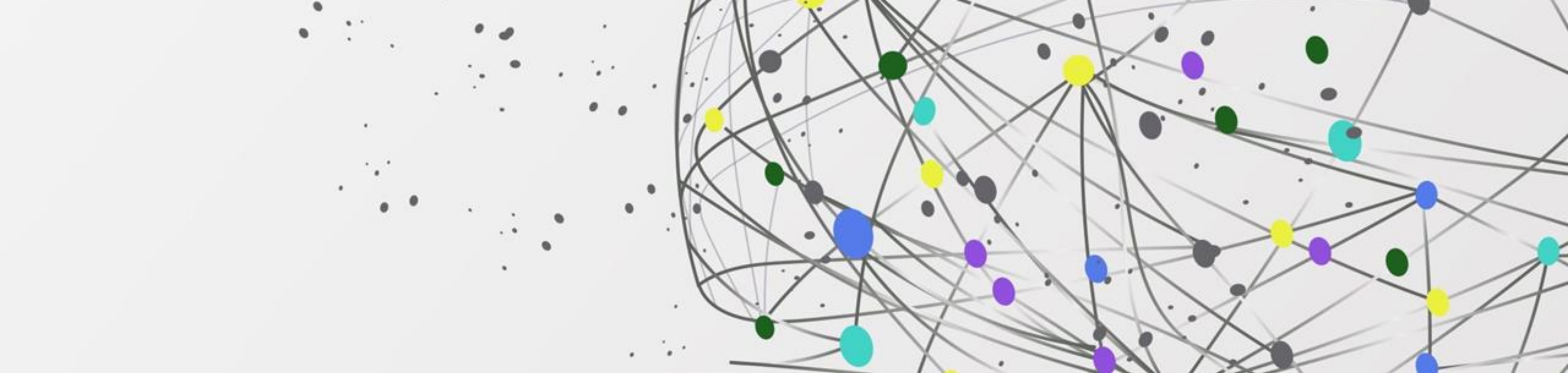

AF35689 – INTERMEDIATE MACROECONOMICS

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Introduction

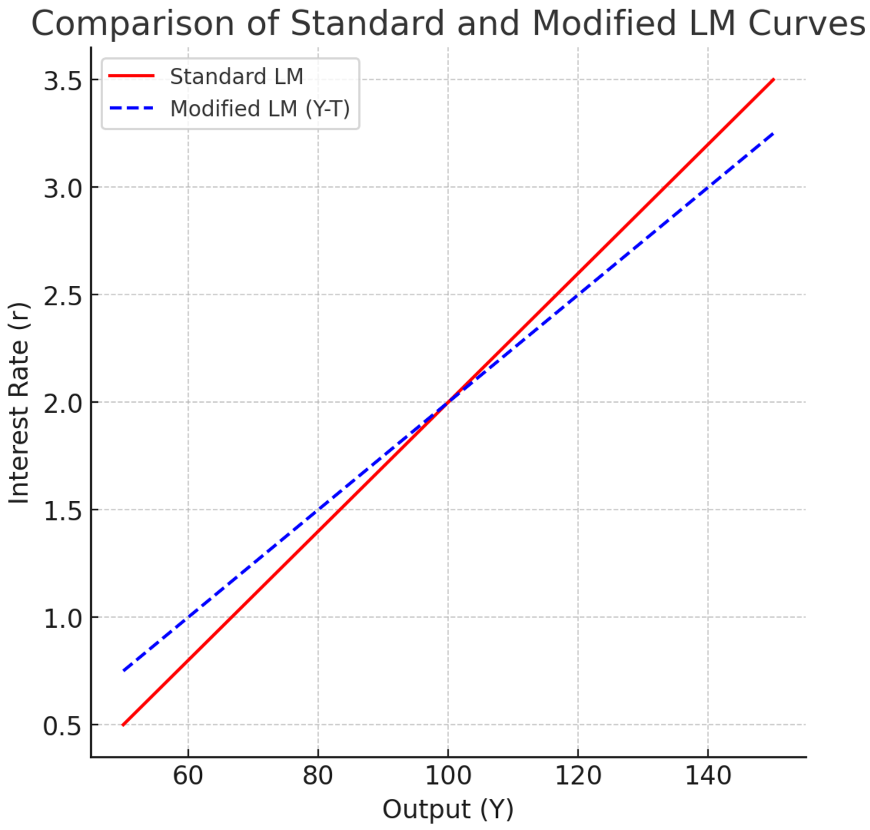
- The IS-LM model explains macroeconomic equilibrium based on the interaction between the goods market (IS curve) and the money market (LM curve).
- In our modification, the money demand function depends on disposable income:

$$\frac{M}{P} = L(r, Y - T)$$

- This affects how fiscal policy (government spending and taxes) influences equilibrium output and interest rates.
- This modification is particularly relevant in economies with significant tax policies affecting disposable income.

Modified Money Demand Function

- Standard LM equation: $\frac{M}{P} = L(r, Y)$
- Modified function: $\frac{M}{P} = L(r, Y - T)$
- Meaning: Money demand depends on after-tax income, affecting liquidity preference.
- This implies that changes in taxation affect not only consumption but also the demand for money, leading to shifts in the LM curve.



IS Curve - Goods Market Equilibrium

- The IS curve represents equilibrium in the goods market:

- Government spending and taxes influence aggregate demand.
- A rise in increases output by boosting government purchases, shifting the IS curve to the right.
- A decrease in increases disposable income, boosting consumption and shifting the IS curve to the right as well.

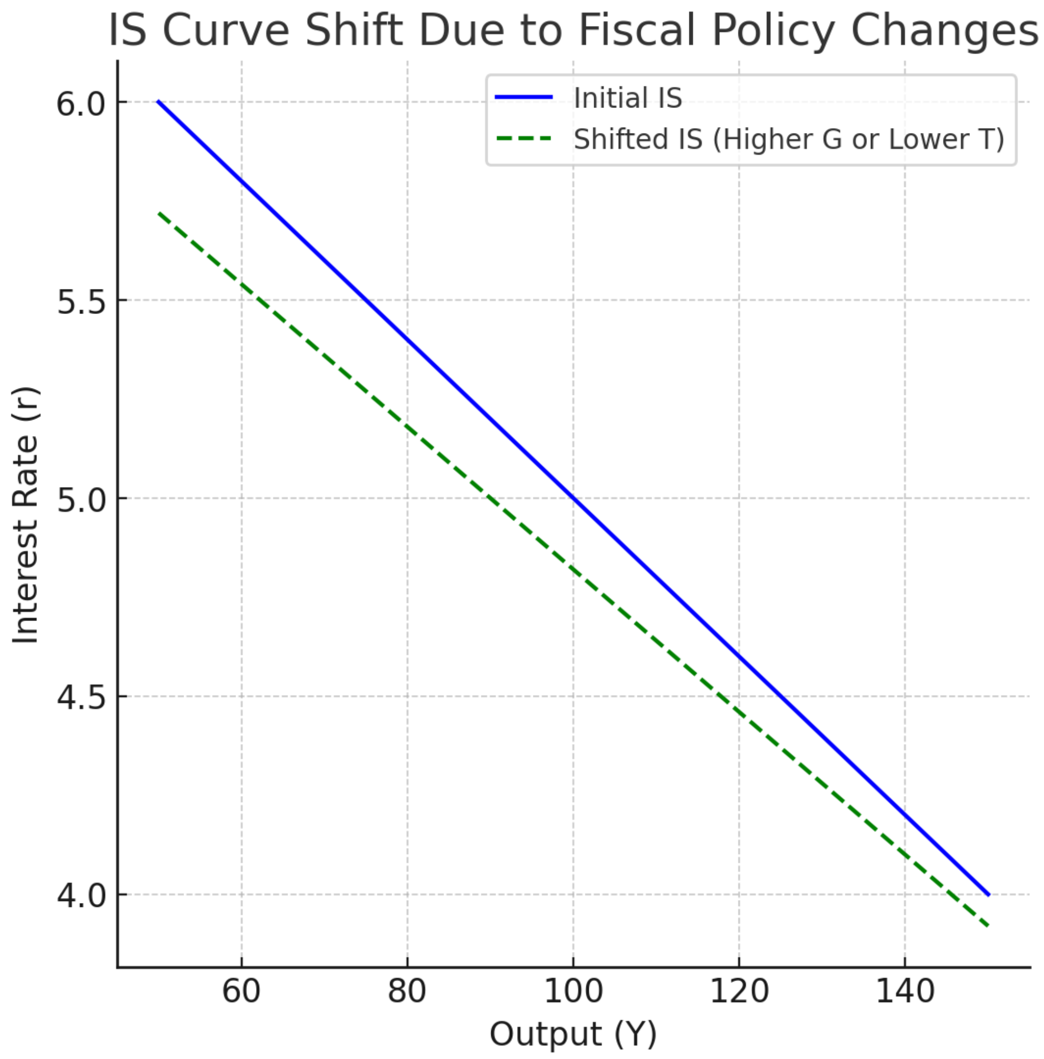
LM Curve – Money Market Equilibrium

- The LM curve represents equilibrium in the money market:

- Since money demand depends on disposable income, tax changes now directly affect the LM curve.
- Lower T increases Y-T , which raises money demand and can lead to higher interest rates.
- Higher T reduces Y-T , leading to lower money demand, which might cause the LM curve to shift downward (lower interest rates).

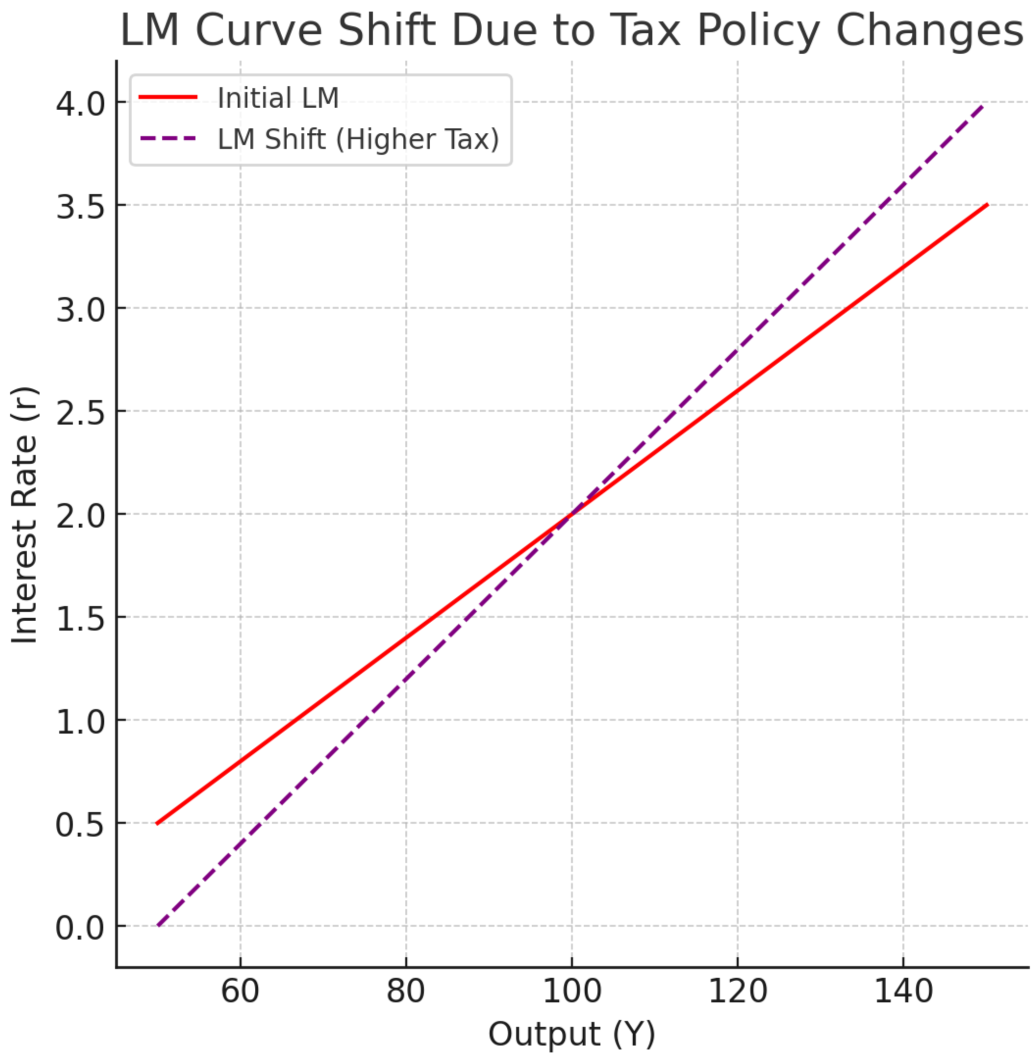
Scenario 1 – Increased Government Spending

- The Bajoran Provisional Government increases G to rebuild after the Cardassian occupation.
- This shifts the IS curve **rightward**, increasing output Y and interest rates r.
- **New effect:** Higher output raises money demand, but since LM depends on Y-T , the response differs from the standard model.
- If the increase in government spending raises overall income, money demand also rises, leading to a shift in the LM curve.



Scenario 2 – Tax Cuts

- Tax cuts increase disposable income, raising consumption and shifting IS **rightward**.
- Since LM depends on Y-T , lower taxes **increase** money demand, shifting LM **leftward**.
- Effect: Higher interest rates, potential crowding out of investment.
- Unlike government spending increases, tax cuts can influence both IS and LM simultaneously.
- If money supply remains constant, the LM shift leftward could dampen the expansionary effects of tax cuts by raising interest rates.



Policy Implications and Considerations

- The modified IS-LM model suggests that fiscal policies (government spending and tax cuts) impact both the goods market and money market.
- Policymakers must consider the simultaneous shifts in IS and LM to avoid undesired effects such as excessive interest rate increases.
- If tax cuts are implemented without monetary expansion, rising interest rates could hinder private investment.
- Government spending, though expansionary, may require careful management to balance inflation risks.
- This model is useful in understanding economic recovery strategies, such as those for the Bajoran Provisional Government.